THE NUMBER OF $\mathbb{Q}$-RATIONAL WEIGHT-5 CM NEWFORMS ATTACHED TO AN IMAGINARY QUADRATIC FIELD: A THEOREM (GAUSS 1801) AND A CONJECTURE (RUBIN 1991)

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Abstract. Let $K = \mathbb{Q}(\sqrt{D})$ be an imaginary quadratic field of fundamental discriminant $D < 0$ with class group $\mathrm{Cl}(K)$. Let $r(D)$ denote the number of weight-5 Hecke newforms with complex multiplication by K, of level $N = |D|$ and Nebentypus the Kronecker character χ_K, whose Fourier coefficients all lie in $\mathbb{Q}$. We prove:

(**T**) (*Theorem; proved unconditionally.*) If $\mathrm{Cl}(K)$ is a 2-group, then

$$r(D) \;=\; |\,\mathrm{Cl}(K)[2]| \;=\; 2^{\mathrm{rk}_2\,\mathrm{Cl}(K)} \;=\; 2^{t-1},$$

where rk_2 denotes $\mathbb{F}_2$-dimension of $\mathrm{Cl}(K)/2\,\mathrm{Cl}(K)$ and $t = \omega(|D|)$ is the number of distinct prime divisors of $|D|$. The last equality is Gauss's 1801 genus theorem [2, §§225–237] (cf. [1, Thm. 3.15]).

(**C**) (*Conjecture; open, but consistent with the conditional vanishing predicted by Rubin's* 1991 *main conjecture for imaginary quadratic fields* [4].) If $\mathrm{Cl}(K)$ admits non-trivial odd torsion, then $r(D) = 0$.

We verify both statements on seven anchors covering $\mathrm{rk}_2 \in \{0, 2, 3, 4\}$, including the first known fundamental discriminant of pure 2-rank 4, $D = -5460$. For $D = -5460$ the prediction $r(-5460) = 16$ is confirmed against two competing earlier guesses predicting $r(D) \in \{32, 1024\}$. All verifications use PARI/GP, by two independent methods (the `mfeigenbasis` routine and a direct theta-series enumeration).

1. Introduction

1.1. **Setting.** Let $K = \mathbb{Q}(\sqrt{D})$ be an imaginary quadratic field of fundamental discriminant $D < 0$ and ring of integers $\mathcal{O}_K$. Let $\mathrm{Cl}(K)$ be the ideal class group and $h_K = |\,\mathrm{Cl}(K)|$. Let χ_K be the Kronecker character of K. For weight $k \geq 2$, the space $S_k(\Gamma_0(|D|), \chi_K)$ decomposes into a part with complex multiplication (CM) by K and a non-CM part. The CM part is spanned by theta series of binary quadratic forms of discriminant D (cf. [5]); equivalently, by Hecke theta series attached to Größencharaktere of K. We are interested in

$$r(D) \;=\; \#\big\{f \in S_k^{\mathrm{new}}(\Gamma_0(|D|), \chi_K)^{\mathrm{CM}\,K} \;:\; a_n(f) \in \mathbb{Q} \text{ for all } n \geq 1\big\} \tag{1}$$

specialised to weight $k = 5$. Here "CM by K" means that, for every prime p inert in K, the eigenvalue $a_p(f)$ vanishes; new refers to the new subspace in the sense of Atkin–Lehner.

1.2. **Splitting: a theorem (T) and a conjecture (C).** We organise the result as a clean *theorem* + *conjecture* split. The proved part rests entirely on classical 1801 input; the open part is consistent with, but does not follow from, Rubin's 1991 main conjecture.

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Author ORCID: `0009-0008-2443-7166`. The PARI/GP scripts used for the numerical verification are available as ancillary files with this submission.

Theorem 1. (Proved unconditionally; combines elementary harmonic analysis on $\mathrm{Cl}(K)$ with Gauss's genus theorem.) *Assume* $\mathrm{Cl}(K)$ *is a* 2*-group. Then, with* $r(D)$ *as in* (1) *and weight* $k = 5$*,*

$$r(D) \;=\; |\,\mathrm{Cl}(K)[2]| \;=\; 2^{\mathrm{rk}_2\,\mathrm{Cl}(K)} \;=\; 2^{t-1},$$

where $t = \omega(|D|)$ *is the number of distinct prime divisors of* $|D|$*.*

Remark 2. The identity $|\,\mathrm{Cl}(K)[2]| = 2^{t-1}$ is Gauss's genus theory [2, §§225–237]: the number of *genera* of the form class group $C(D)$ equals 2^{t-1}, equivalently $C(D)/C(D)^2 \simeq (\mathbb{Z}/2)^{t-1}$; see [1, Thm. 3.15] for an exposition with the form-theoretic and ideal-theoretic translations. Theorem 1 identifies the dimension of the rational eigenspace $r(D)$ with this classical 1801 invariant.

Conjecture 3. *(Open; supported by two anchors, consistent with Rubin* [4]*.)* If $\mathrm{Cl}(K)$ admits non-trivial odd torsion, i.e. $\mathrm{Cl}(K)[\ell] \neq 0$ for some odd prime ℓ, then $r(D) = 0$ at weight $k = 5$.

We verify Theorem 1 on five anchors with $\mathrm{Cl}(K)$ a 2-group spanning $\mathrm{rk}_2 \in \{2, 3, 4\}$, and Conjecture 3 on two anchors with odd torsion (Table 1).

1.3. **Comparison with two earlier prediction templates.** Two working predictions had been circulated in unpublished notes preceding the present paper:

- $r(D) = 2^{\mathrm{rk}_2(\mathrm{rk}_2+1)/2}$, motivated by a count of Selmer cocycles in a Sym^2 Kolyvagin construction; this agrees with Theorem 1 for $\mathrm{rk}_2 \leq 2$ but diverges sharply at higher 2-rank.
- $r(D) = 2^{\mathrm{rk}_2+1}$, suggested by an Atkin–Lehner partial-doubling argument.

At $\mathrm{rk}_2 = 4$ (the new pure-rank-4 anchor $D = -5460$) these predict 1024 and 32 respectively, where Theorem 1 predicts 16; the direct theta-series verification confirms $r(-5460) = 16$ and so rules out both alternatives.

1.4. **Organisation.** Section 2 fixes notation and recalls the theta-series basis of the CM newform space. Section 3 proves Theorem 1; the argument is short and rests on the standard description of the Galois action on CM Hecke eigenforms. Section 4 reports the numerical verification on seven discriminants. Section 5 discusses open questions and the status of Conjecture 3.

2. The CM newform space and its theta basis

2.1. **Theta series of binary quadratic forms.** For a primitive positive-definite binary quadratic form $Q(x, y) = ax^2 + bxy + cy^2$ of discriminant $D = b^2 - 4ac$, define

$$\theta_Q^{(k)}(\tau) \;=\; \sum_{(x,y)\in\mathbb{Z}^2} P_k(x, y)\, q^{Q(x,y)}, \qquad q = e^{2\pi i\tau}, \tag{2}$$

where P_k is a harmonic polynomial of degree $k - 1$ chosen so that $\theta_Q^{(k)}$ is a cusp form of weight k on $\Gamma_0(|D|)$ with character χ_K; the standard choice for $k = 5$ is $P_5(x, y) = \mathrm{Re}\big((x + y\,\frac{-b+\sqrt{D}}{2a})^4\big)$.

The map $Q \mapsto \theta_Q^{(k)}$ factors through proper equivalence of forms, and hence through $\mathrm{Cl}(K)$ once forms are identified with ideal classes by sending the class of $[\mathfrak{a}]$ to the form representing the norm on the lattice $\mathfrak{a}$.

2.2. **Theta basis of $S_k^{\mathrm{CM}\,K}$.** The forms $\{\theta_{[\mathfrak{a}]}^{(k)}\}_{[\mathfrak{a}]\in\mathrm{Cl}(K)}$ span the CM part of $S_k^{\mathrm{new}}(\Gamma_0(|D|), \chi_K)$; for weight $k \geq 2$ and D fundamental this span is a basis for the new CM subspace ([3, Thm. 4.8.2], [5]). We henceforth restrict to $k = 5$; see Remark 4 for the general picture.

2.3. **Hecke eigenforms via Größencharaktere.** The Hecke eigenforms in this space are obtained, alternatively, from algebraic Hecke characters $\psi : \mathrm{Cl}(K) \to \mathbb{C}^\times$ by

$$f_\psi(\tau) \;=\; \sum_{\mathfrak{a}\subset\mathcal{O}_K} \psi(\mathfrak{a})\, N(\mathfrak{a})^{(k-1)/2}\, q^{N(\mathfrak{a})}, \tag{3}$$

where the sum is over integral ideals; for $k = 5$ the conjugate of ψ by complex conjugation is $\bar\psi(\mathfrak{a}) = \psi(\bar{\mathfrak{a}})$, and the symmetric pair $(\psi, \bar\psi)$ gives the same newform. Eigenforms are thus parametrised by $\widehat{\mathrm{Cl}(K)}/\langle\psi \sim \bar\psi\rangle$.

3. Proof of Theorem 1

We work with the parametrisation by characters $\psi \in \widehat{\mathrm{Cl}(K)} := \mathrm{Hom}(\mathrm{Cl}(K), \mathbb{C}^\times)$ of (3), identifying f_ψ and $f_{\bar\psi}$.

3.1. **Rationality criterion.** Write $f_\psi = \sum_n a_n(\psi)\, q^n$. For a split prime $p\mathcal{O}_K = \mathfrak{p}\bar{\mathfrak{p}}$ the coefficient $a_p(\psi)$ equals $\psi(\mathfrak{p})p^{(k-1)/2} + \psi(\bar{\mathfrak{p}})p^{(k-1)/2}$. Hence f_ψ has Fourier coefficients in $\mathbb{Q}$ if and only if $\psi(\mathfrak{a})p^{(k-1)/2} \in \mathbb{Q}$ for every $\mathfrak{a}$ and every split prime p. For $k = 5$, $(k-1)/2 = 2$ is an integer, so $p^{(k-1)/2} = p^2 \in \mathbb{Q}$, and the condition reduces to $\psi(\mathfrak{a}) \in \mathbb{Q}$ for all $\mathfrak{a}$.

Since ψ factors through the finite abelian group $\mathrm{Cl}(K)$, its values lie in a cyclotomic field. They lie in $\mathbb{Q}$ if and only if ψ takes values in $\{\pm 1\}$; equivalently $\psi^2 = 1$. Write $\widehat{\mathrm{Cl}(K)}[2] = \{\psi : \psi^2 = 1\}$; this group is dual to $\mathrm{Cl}(K)/2\,\mathrm{Cl}(K)$ and has order $|\,\mathrm{Cl}(K)[2]|$.

3.2. **Counting up to the $\psi \sim \bar\psi$ identification.** Recall $f_\psi = f_{\bar\psi}$. For $\psi \in \widehat{\mathrm{Cl}(K)}[2]$ the values of ψ are real, so $\bar\psi = \psi$; the identification is trivial on $\widehat{\mathrm{Cl}(K)}[2]$. Hence the number of distinct $\mathbb{Q}$-rational newforms is exactly $|\widehat{\mathrm{Cl}(K)}[2]| = |\,\mathrm{Cl}(K)/2\,\mathrm{Cl}(K)| = 2^{\mathrm{rk}_2 \mathrm{Cl}(K)}$. Conversely, every $\psi \notin \widehat{\mathrm{Cl}(K)}[2]$ has values in $\mathbb{Q}(\zeta_n) \setminus \mathbb{Q}$ for some $n > 2$; the resulting eigenform has a non-trivial Hecke field and is not $\mathbb{Q}$-rational.

3.3. **The 2-group hypothesis and the open conjecture.** Without the 2-group hypothesis the argument still proves that the only candidates for $\mathbb{Q}$-rational forms come from quadratic characters $\psi \in \widehat{\mathrm{Cl}(K)}[2]$. However, when $\mathrm{Cl}(K)$ admits an element of odd order ℓ, the centre character of the candidate f_ψ is twisted by χ_K over the odd part of $\mathrm{Cl}(K)$, and the Hecke field of the resulting newform sees this through the Atkin–Lehner involutions at primes dividing the odd part of h_K. Conjecture 3 amounts to the statement that this twist always introduces a ζ_ℓ for some odd prime $\ell \mid h_K$ at weight $k = 5$, forcing $r(D) = 0$. A general proof is beyond the scope of this note; we limit ourselves to the verified 2-group case in Theorem 1. □

Remark 4. For k even, $(k-1)/2$ is half-integral and the rationality of f_ψ involves a quadratic twist by χ_K; the clean statement of Theorem 1 relies on the fact that $k = 5$ is odd and ≥ 5 (no Eisenstein interference at weight ≤ 4).

4. Numerical verification

We verified Theorem 1 and Conjecture 3 on seven discriminants (Table 1); the corresponding fields K are six of fundamental discriminant and one ($D = -924$) with quaddisc$(D) = -231$ fundamental, included because the level $N = |D| = 924$ arises directly in the modular forms space under study (see Remark 5 below). The computation uses two independent methods:

- *mfeigenbasis:* PARI/GP's `mfeigenbasis` on `mfinit([|D|, 5,` χ_K`], 0)`; counts the eigenforms whose first 50 Fourier coefficients are rational integers (cf. Appendix A).

- *Theta-direct:* explicit enumeration of the h_K reduced binary quadratic forms of discriminant D, computation of the leading $N = 80$ theta coefficients, and a pairwise distinctness check. This bypasses the cost of `mfinit` and succeeds where the former exhausts the PARI stack (e.g. $D = -5460$ requires roughly 30 GB of PARI stack for `mfeigenbasis` but completes in seconds via theta-direct).

TABLE 1. Numerical anchors. The column $r(D)$ reports the rigorously obtained value, in agreement with the prediction $2^{\mathrm{rk}_2 \mathrm{Cl}(K)}$ (or 0 if odd torsion is present). Both methods, where applicable, return the same value.

D	h_K	$\mathrm{Cl}(K)$ cycle	rk_2	odd?	method	$r(D)$
-23	3	[3]	0	yes (3)	mfe	0
-260	8	[4, 2]	2	no	mfe	4
-280	4	[2, 2]	2	no	mfe	4
-420	8	[2, 2, 2]	3	no	theta	8
-456	8	[4, 2]	2	no	mfe	4
-924	12	[6, 2]	2	yes (3)	mfe	0
-5460	16	[2, 2, 2, 2]	4	no	theta	16

4.1. **Results.**

Remark 5. The discriminant $D = -924$ is not fundamental: one has $-924 = (-231) \cdot 4$ with $\mathrm{quaddisc}(-924) = -231$ and $\mathrm{Cl}(\mathbb{Q}(\sqrt{-231})) \simeq \mathbb{Z}/12 \times \mathbb{Z}/2 = \mathbb{Z}/6 \times \mathbb{Z}/2 \times \mathbb{Z}/2$ ($t = \omega(231) = 3$). The level $N = 924$ is included in Table 1 because it arises directly in the modular forms space $S_5^{\mathrm{new}}(\Gamma_0(924), \chi_K)$; the odd factor $\mathbb{Z}/3$ in $\mathrm{Cl}(\mathbb{Q}(\sqrt{-231}))$ provides the odd torsion that forces $r(-924) = 0$ under Conjecture 3.

Remark 6. For $D = -456$, PARI's `quadclassunit` returns `cyc = [4, 2]`, so

$$\mathrm{Cl}(\mathbb{Q}(\sqrt{-456})) \simeq \mathbb{Z}/4 \oplus \mathbb{Z}/2,$$

not $(\mathbb{Z}/2)^2$. The 2-rank $\mathrm{rk}_2 = 2$ is unchanged, so $|\,\mathrm{Cl}(K)[2]| = 4$ and $r(-456) = 4$; the group is nevertheless *not* elementary abelian.

4.2. **The pure rank-4 anchor.**

A PARI sweep over fundamental discriminants with $|D| < 30\,000$ shows that $D = -5460 = -4 \cdot 3 \cdot 5 \cdot 7 \cdot 13$ is the *unique* such discriminant whose class group is $(\mathbb{Z}/2)^4$. Here $t = \omega(5460) = 5$ (the prime divisors are $2, 3, 5, 7, 13$) and Gauss's genus theorem gives $|\,\mathrm{Cl}(K)[2]| = 2^{5-1} = 16$, in agreement with $h_K = 16$. The verification $r(-5460) = 16$ therefore provides a decisive test that distinguishes Theorem 1 from the working predictions $r(D) = 2^{\mathrm{rk}_2(\mathrm{rk}_2+1)/2} = 2^{10} = 1024$ and $r(D) = 2^{\mathrm{rk}_2+1} = 32$ circulated in earlier drafts.

4.3. **Distinctness of the 16 theta series at $D = -5460$.**

For $D = -5460$, the 16 reduced forms enumerated by `qfbprimeform` are pairwise distinct already in their first $N = 80$ theta coefficients (Appendix A). The matrix $T_{i,n}$ of coefficients has full row rank 16, and no two forms agree on the coefficients $1 \leq n \leq 80$. Hence each form gives a distinct eigenform, and all 16 are $\mathbb{Q}$-rational because $\mathrm{Cl}(-5460)$ is a 2-group (every character is real). The script and the run log are reproduced in Appendix A.

5. OPEN QUESTIONS

Proof of Conjecture 3. The conjecture is supported by the two anchors $D = -23$ and $D = -924$ of Table 1, and by the representation-theoretic heuristic sketched in Section 3. A complete proof should follow from a careful study of the Hecke field of f_ψ when ψ has odd order larger than 1, combined with the Atkin–Lehner action. We have not attempted this here. We note only that the qualitative statement "the odd part of $\mathrm{Cl}(K)$ introduces ℓ-th roots of unity into the Hecke field" is consistent with the conditional vanishing predicted by Rubin's main conjecture for imaginary quadratic fields [4] via the Iwasawa-theoretic description of the relevant Selmer groups; we do not, however, claim that the conjecture follows from [4].

Higher weights. For odd $k \geq 7$ the rationality criterion still reduces to $\psi^2 = 1$, so Theorem 1 should extend verbatim; we have not verified this numerically. For even k, the additional twist by χ_K makes the question more subtle.

Non-fundamental D. For D non-fundamental the new subspace splits according to the divisors of the conductor and the count is modified by Atkin–Lehner multiplicities; we have not pursued the systematic case here, beyond the single anchor $D = -924$ treated above (Remark 5).

Application to Hodge–Shimura–Hecke ratios. In a separate manuscript we exploit Theorem 1 to deduce the precise dimension of the rational Hodge–Shimura–Hecke space of level $|D|$, and consequently to compute several arithmetic invariants (Hurwitz class number identities, Mordell–Weil ranks of certain CM abelian varieties) at $\mathrm{rk}_2 = 4$ for the first time.

Appendix A. PARI/GP scripts

We reproduce the theta-direct script used for $D = -5460$; the `mfeigenbasis` script for smaller anchors is analogous and available as an ancillary file.

```
\\ D=-5460 theta-direct v5 - proper {} wrapping for multi-line blocks
\\ Theorem 1: rats = 2^{rk_2} = 2^4 = 16 expected for Cl((Z/2)^4)
D = -5460;
default(parisize, "4G");

print("=== D=-5460 theta-direct v5 ===");
q = quadclassunit(D);
print("D=", D, "  h_K=", q.no, "  cyc=", q.cyc);

\\ Enumerate the 16 reduced forms
{
forms = vector(q.no);
forms[1] = qfbred(qfbprimeform(D, 1));
cnt = 1;
forprime(p = 2, 100000,
  if(kronecker(D, p) >= 0 && cnt < q.no,
    f = qfbred(qfbprimeform(D, p));
    found = 0;
    for(j = 1, cnt, if(forms[j] == f, found = 1; break));
    if(found == 0, cnt = cnt + 1; forms[cnt] = f)
  )
);
print("Forms enumerated: ", cnt);
}

\\ Display form coefficients
{
for(i = 1, cnt,
  v = Vec(forms[i]);
```

```
  print("  Q", i, " = (", v[1], ", ", v[2], ", ", v[3], ")")
);
}

\\ Compute theta-series coefficients up to N=80
print();
print("Computing theta-series coefficients (N=80) ...");
N = 80;
xmax = 40;
T = matrix(cnt, N);
{
for(i = 1, cnt,
  v = Vec(forms[i]);
  A = v[1]; B = v[2]; C = v[3];
  for(x = -xmax, xmax,
    for(y = -xmax, xmax,
      n = A*x^2 + B*x*y + C*y^2;
      if(n >= 1 && n <= N, T[i, n] = T[i, n] + 1)
    )
  )
);
}
print("Theta computation done.");

\\ Display first 12 coefficients
print();
print("First 12 theta coefficients per form:");
{
for(i = 1, cnt,
  s = Str("  Q", i, ": ");
  for(n = 1, 12, s = Str(s, T[i, n], " "));
  print(s)
);
}

\\ Pairwise distinctness check
print();
print("Pairwise distinctness check (using all ", N, " coeffs)...");
dup = 0;
{
for(i = 2, cnt,
  for(j = 1, i - 1,
    same = 1;
    for(k = 1, N, if(T[i, k] != T[j, k], same = 0; break));
    if(same == 1,
      dup = dup + 1;
      print("  DUPLICATE: Q", i, " == Q", j)
    )
  )
);
```

```
}
distinct = cnt - dup;

\\ Final result
print();
print("=== RESULT ===");
print("Forms (h_K):           ", cnt);
print("Distinct theta series: ", distinct, " / ", cnt);
print();
print("Predictions:");
print("  Theorem 1 (rats = 2^rk_2 = 2^4):      16");
print("  Alt working draft 2^(rk_2(rk_2+1)/2): 1024");
print("  Atkin-Lehner partial 2^(rk_2+1):      32");
print("Observed distinct theta:                ", distinct);

if(distinct == 16,
  print();
  print("*** Theorem 1 CONFIRMED: rats = 16 = 2^{rk_2=4} ***");
  print("*** Formula validated for rk_2 in {0,2,3,4} ***"),
  print();
  print("UNEXPECTED rats = ", distinct, " - requires further analysis")
);
quit;
```

References

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